

Make every figure earn its place

Repeatable methods • captions that stand alone • writing about figures

YOU LEAVE WITH

A peer-checked Methods section, report-ready captions and one result sentence per figure.



Tutorial 5 of 8 • Semester 1

Today's plan

Goal: Leave with report-ready captions and a Methods section someone else could repeat.

- 1 Peer check: could I repeat your Methods?
- 2 Methods: r isn't enough (coins, n , p , n_{eff})
- 3 Methods: noise maps, year-to-year tests, thresholds
- 4 Captions, writing about figures, honest colour scales

BRING / HAVE READY

- Task 4B Methods draft (printed or on screen)
- Task 4C topic-sentence outline
- Your 3–5 draft figures



IN-CLASS ACTIVITY • 10 MIN

Could I repeat your study?

- 1 Swap Methods drafts with a partner.
- 2 Read as if you had to **redo the analysis tomorrow** with only this text and the notebook.
- 3 List every decision you would have to **guess**: period? region bounds? ONI threshold? anomaly baseline? weighting? cloud variable? months used?
- 4 Also mark anything that is **code, not science** (cell numbers, function names, 'I ran...').
- 5 Return the list. Don't fix it now; that is Task 5A.

YOU LEAVE WITH

- A 'guess list' for your own Methods
- A sense of what a complete Methods looks like

THE STANDARD

- Schultz: a tenet of scientific writing is whether a reader 'has enough information from the paper to duplicate the study'.
- Interpretation (why the cloud changes) belongs in the **Discussion**, not the Methods.

METHODS

Coins in a bucket: r is not enough, n matters

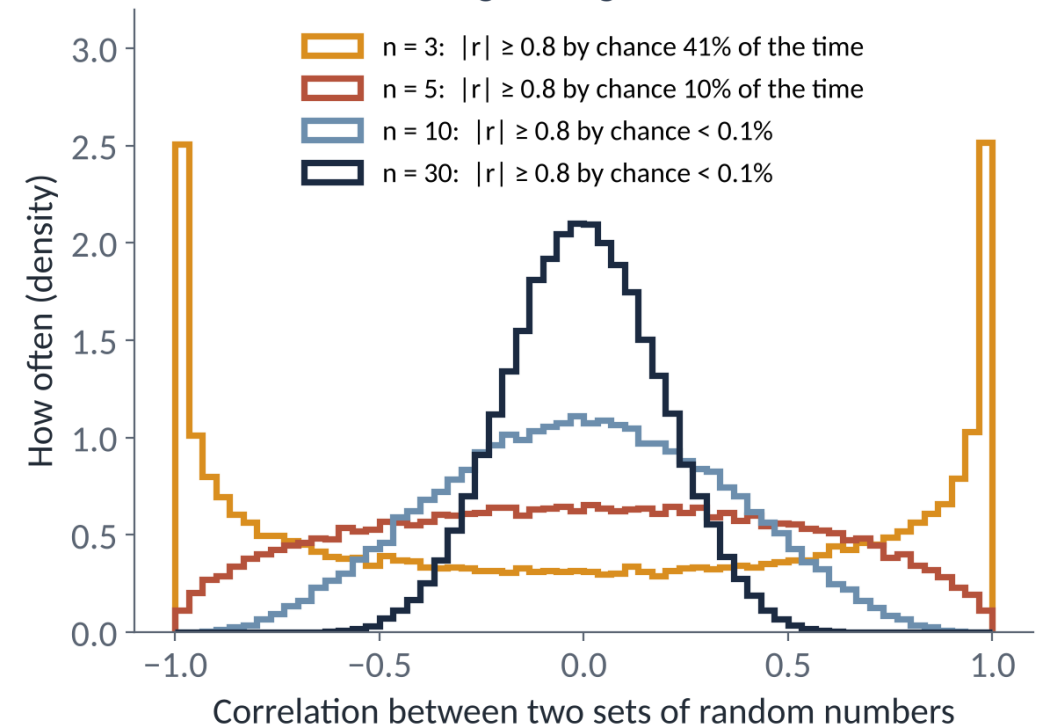


Pull them out at random. What's the chance they come out in perfect order, so that draw order and value have $r = 1$?

3 coins	1 in 6	17%
5 coins	1 in 120	0.8%
7 coins	1 in 5,040	0.02%

THE POINT

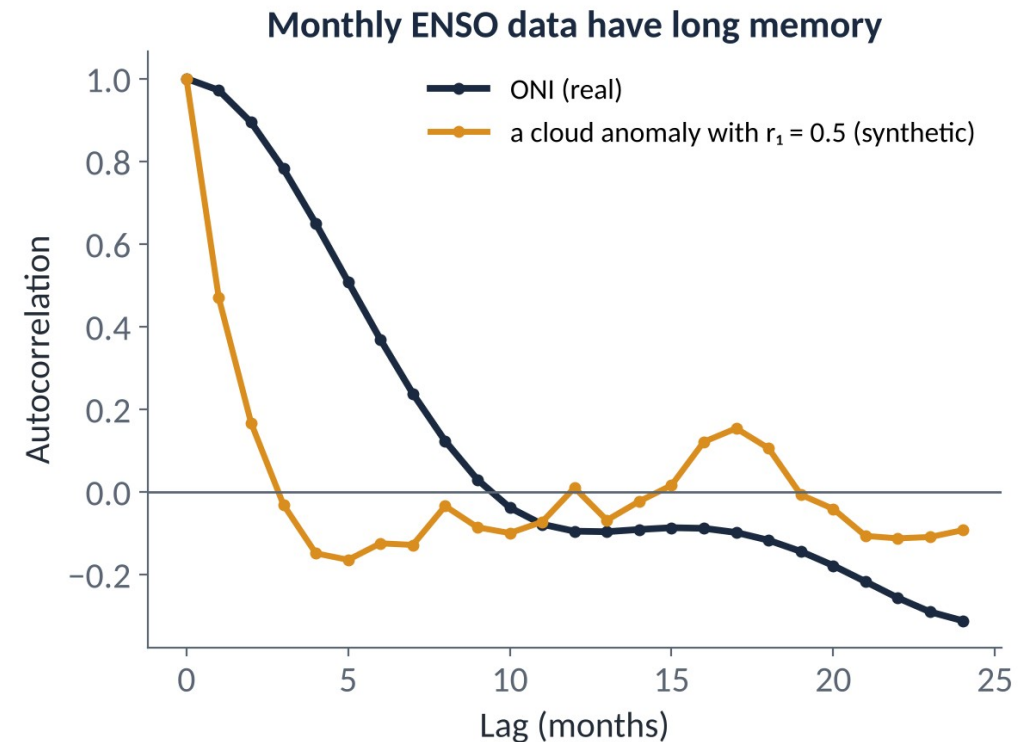
- The **same** perfect r , but a very different chance that it's luck. The p-value combines r **and** n .
- $|r|$ needed for $p < 0.05$: $n = 10 \rightarrow 0.63 \cdot n = 30 \rightarrow 0.36 \cdot n = 85 \rightarrow 0.21 \cdot n = 247 \rightarrow 0.12$

Pure chance gives big r when n is small

METHODS

Are 247 months independent? Effective n

- A p-value assumes **independent** samples. Monthly ENSO data aren't: the ONI's lag-1 autocorrelation is **0.97**.
- **Effective sample size:** $n_{\text{eff}} = n (1 - r_1) / (1 + r_1)$, where r_1 is each series' lag-1 autocorrelation.
- With $n = 247$ months, $n_{\text{eff}} \approx$ **135, 85 or 31** if the cloud anomaly's r_1 is 0.3, 0.5 or 0.8.
- Notebook §6 now prints **n and n_eff**; its p_{adj} and 95% range use n_{eff} . Report those.

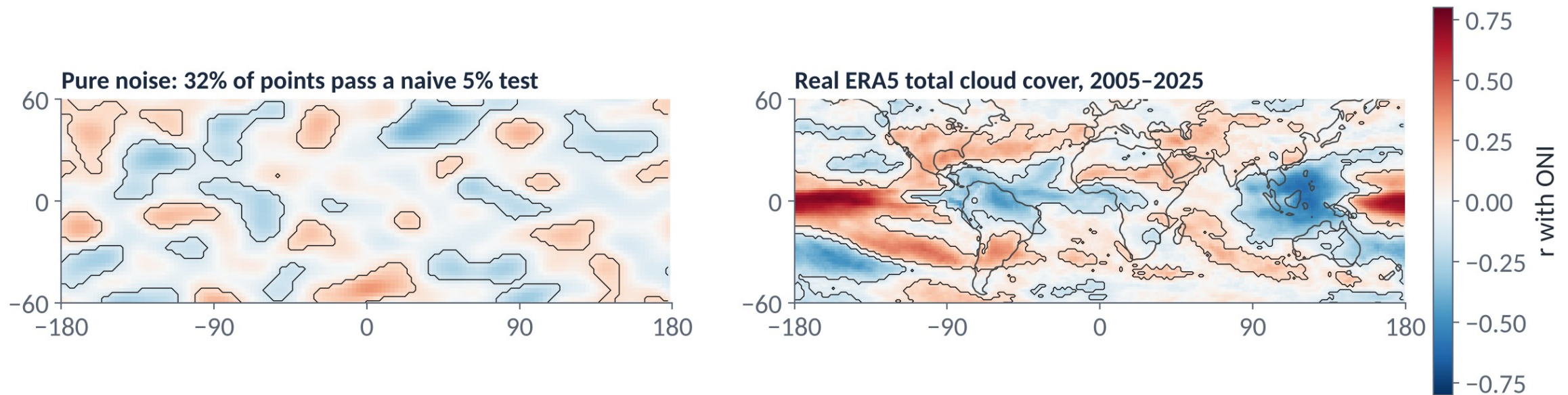


CALCULATE IT YOURSELF

$t = r \sqrt{(n - 2) / (1 - r^2)}$, with $n - 2$ degrees of freedom. Task 5D: compute p with n , then with n_{eff} .

METHODS

Pure noise can draw convincing maps

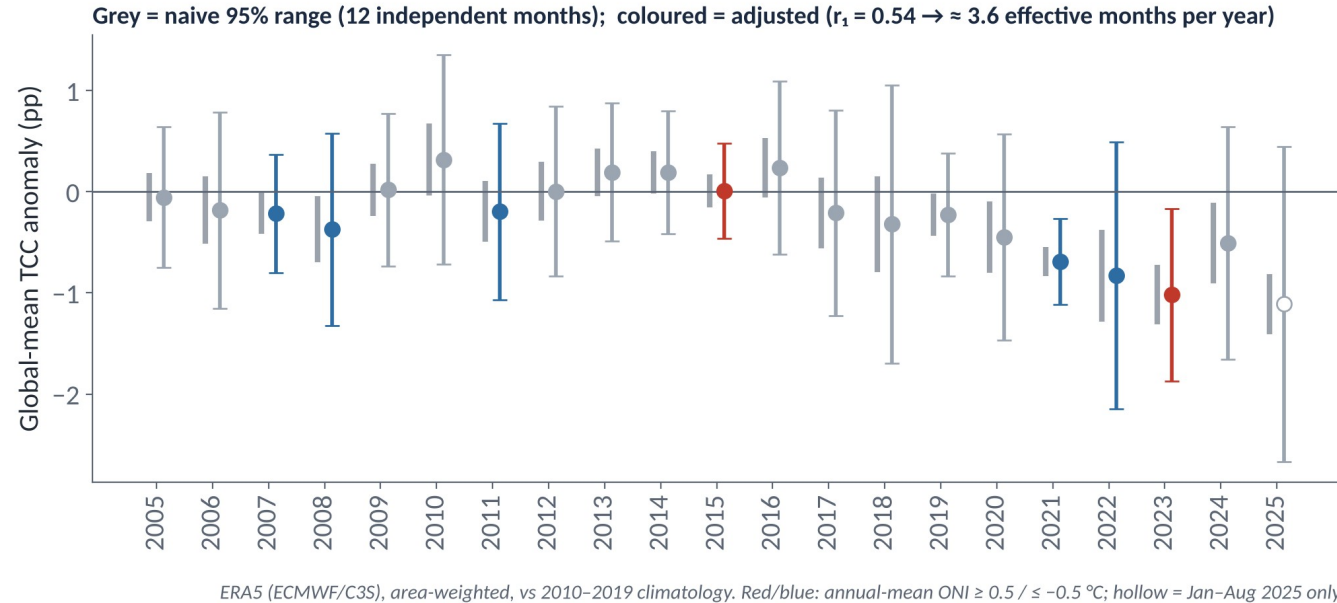


Both use the same real ONI ($n = 247$ months). Thin black contours: $|r| > 0.125$ (naive 5% test using n , not n_{eff}). Real panel: ERA5 (ECMWF/C3S) anomalies vs 2010–2019 climatology.

- Left: the **real ONI** correlated with **pure noise** that has realistic memory and blobs. **32%** of grid points pass a naive 5% test.
- Even with no memory at all, 5% pass by chance. A map has thousands of points, so that's **hundreds** of false 'signals' (Wilks 2016).
- Right: the real signal is far stronger and sits where the physics says it should. Choose regions for a **physical** reason, then test them properly. Try it: `methods_demos.ipynb §6`.

METHODS

Is one year's cloud different from another's?



- Compare two years' 12 monthly anomalies with a **t-test** $\rightarrow$ p. New notebook section **3b** does it for any two years.
- Global cloud anomalies have $r_1 = 0.54$, so 12 months $\approx$ **3.6** independent months. **2010 vs 2011**: naive $p = 0.02$, adjusted $p = 0.24$.
- **190 pairs** of complete years $\rightarrow$ ~ 10 'significant' by chance. Choose years **before** looking, for a reason.
- Global cloud falls by about **0.4 pp per decade** here. Is that real, or a reanalysis artefact? (Cf. record-low albedo in 2023: Goessling et al. 2024.) Trends can masquerade as ENSO signals. Try DETREND = True.

IN YOUR RESULTS, THIS BECOMES

"Global-mean TCC was 0.51 pp higher in 2010 than in 2011, but the difference is not significant once autocorrelation is allowed for ($p = 0.24$)."

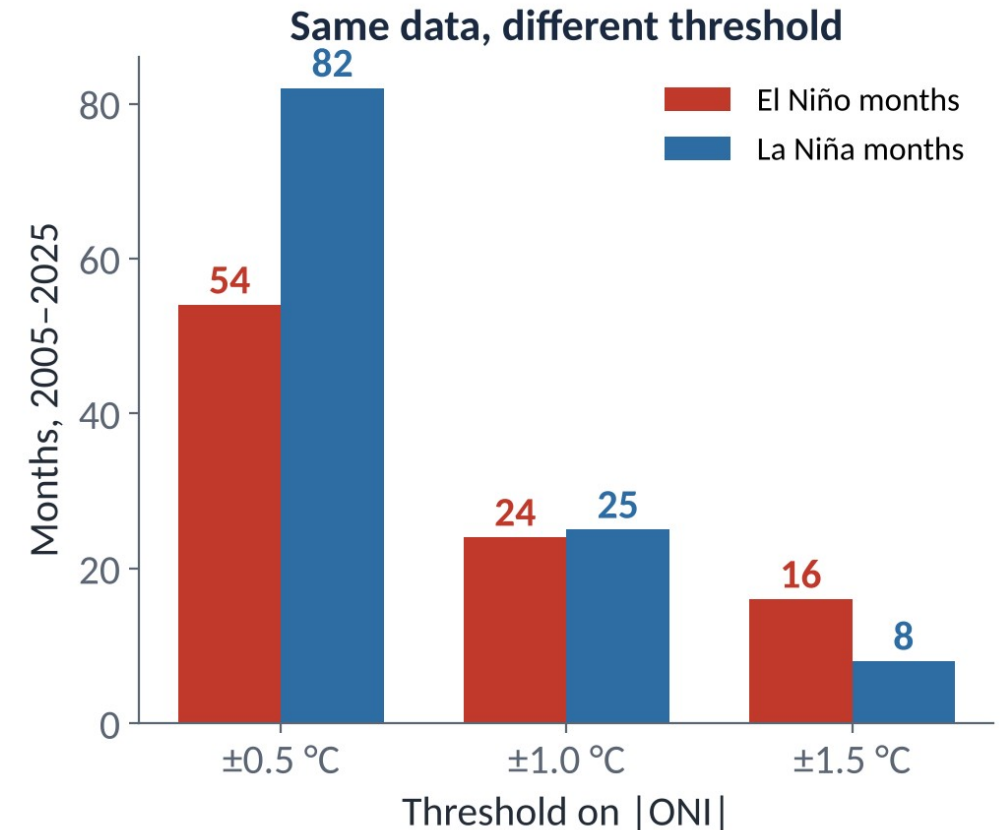
METHODS

Defend your threshold, and test it

- **Schultz:** reviewers will question your thresholds. Defend them by **convention or physics** (NOAA's ± 0.5 °C) or by the **data** (e.g. top and bottom 25% of ONI months).
- Raising the threshold to ± 1.0 °C cuts El Niño months from **54 to 24** and La Niña from **82 to 25**: stronger events, fewer samples.
- **Sensitivity test:** does your answer survive ± 1.0 °C, or leaving out 2015–16? One sentence in Results is the ‘careful treatment of a subtlety’ in the 80%+ band.
- **Composites:** check that most members show the feature, not just one big event (§18.4.4).

ONE SENTENCE IN YOUR RESULTS

“The sign of the composite is unchanged with a ± 1.0 °C threshold (24 El Niño, 25 La Niña months).”



A caption that stands alone

WEAK

“Figure 2. Correlation map.”

STRONG

“Figure 2. Correlation between monthly high-cloud-cover anomaly and the ONI at each grid point, January 2005 – July 2025 (n = 247 months). Boxes mark the central Pacific (CP) and Maritime Continent (MC) regions analysed in Figs 3–4. Figure produced by the author using ERA5 (ECMWF / Copernicus Climate Change Service) and ONI (NOAA Climate Prediction Center) data.”

WHAT CHANGED

- **Essence first** (‘Correlation between ... and the ONI’), then the period, the regions and links to later figures.
- Defines what a **positive value** means implicitly: higher cloud anomaly when ONI is higher.
- Attribution: **your** figure, **their** data. Longer, but every word is doing a job.

Describe the science, not the plot

Schultz's example: "Figure 5 shows plots of surface temperature... showing that..." → "The elimination of sea ice... resulted in warmer arctic temperatures (Fig. 5)."

WEAK

"Figure 3 shows the time series of low-cloud anomaly and ONI. It can be seen from Figure 3 that there is a relationship between them."

STRONG

"High-cloud cover over CP is higher than usual during El Niño and lower during La Niña (Fig. 3); the largest positive anomaly, [] percentage points, coincides with the 2015–16 El Niño." (illustrative)

WHAT CHANGED

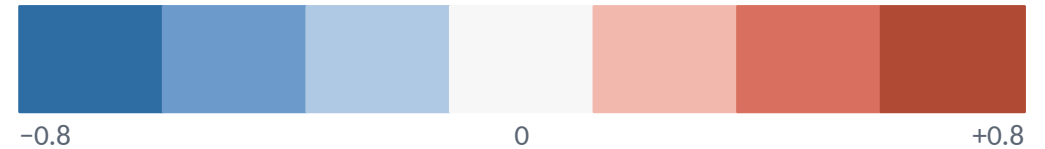
- The **science** is the subject of the sentence; the figure is cited **indirectly**, in brackets, at the end.
- Note the **obvious first**, then the finer point or exception (Schultz §11.11). Don't explain how to read the plot: the caption does that.
- Every figure is cited in the text, **in numerical order**. The text should make sense even without the figures.

METHODS

Honest colour scales

- **Anomalies, differences and correlations:** use a **diverging** colour map (blue–white–red) **centred on zero**, with symmetric limits (e.g. -0.8 to $+0.8$).
- Panels you compare must share the **same** limits, or the eye compares colours that mean different things.
- Avoid rainbow ('jet') maps: they invent sharp boundaries and fail for colour-blind readers.
- Label the colour bar with the **variable and units**.

✓ DIVERGING, CENTRED ON ZERO



✗ RAINBOW: FALSE BOUNDARIES



✗ NOT CENTRED: zero looks 'positive'



IN-CLASS ACTIVITY • 10 MIN

Caption clinic

- 1 Write a **full caption** for each figure: what is plotted • units • period/region • threshold/definition • data source. Begin with the **essence** phrase.
- 2 Under each caption, write the **one result sentence** that will cite it indirectly: science first, “(Fig. n)” last.
- 3 Swap **captions only** (no figures!) with a partner.
- 4 Partner **sketches** what they think each figure looks like: axes, map, colours, panels.
- 5 Compare the sketch with the real figure. Where they differ, the caption is incomplete.

YOU LEAVE WITH

- Report-ready captions for all figures (Task 5C)
- One result sentence per figure: the seeds of your Results

PAIR CHECK

- Is each figure's **essence phrase** different from the others?
- Is the **sign convention** clear (what does positive mean)?
- Is the **attribution** line there?

Before Tutorial 6 • Thu 19 Nov

One week. About 2½ hours. This is the week your Results section appears.

5A • 30 min

Revise Methods

- Fix every item on your partner's guess list
- Remove any code or interpretation
- Stay within ~300 words

5B • 90 min

Results draft, ~550–650 words

- **One paragraph per claim** from your storyboard
- Topic sentence = the finding; then quantified evidence
- Numbers with **units, baseline and sample size**

5C • 20 min

Final captions

- All 3–5 figures, one consistent format
- Five-point check + attribution
- Numbered in the order you cite them

5D • 20 min

Calculate a p-value

- Take **r** and **n** for your region from the \$5 panel
- Compute t and p (workbook / methods_demos \$4)
- Repeat with **n_eff** from \$6. Which do you report?

NOTEBOOK STRAND

- Finalise 3–5 figures (labels, units, centred colour scales); **save them outside Binder**. Optional: a ± 1.0 °C sensitivity test.
- **methods_demos.ipynb \$3–6**; the shift test (\$7) for your region; new **\$3b** if you compare years.

READ • SCHULTZ (2009) ELOQUENT SCIENCE

- **\$11.10–11.14** Captions; discussing and citing figures
- **\$10.1** Concision (and Table 10.2)